

Emerging Hotspots in the U.S. Real Estate Market: A Data-Driven Approach to Investment Opportunities

Tufts University

DATA-200

Deepika Asandi, Jackson Reydel, Amos Trinidad Gonzalez

December 2023

1. Introduction

In the realm of real estate, "hotspots" refer to geographical areas experiencing a rapid increase in investment interest, often due to factors such as economic growth, demographic shifts, or attractive property values. These areas are typically characterized by a high potential for returns on investment. This makes them particularly appealing to investors and real estate professionals.

Our study aims to identify and analyze these hotspots within the U.S. real estate market. This research analysis leverages advanced data analytics, including clustering techniques, and linear and logistic regression models to examine key variables like housing prices, days on Zillow, sale prices, and demographic features. A key criterion for an ideal hotspot in our dataset is properties remaining on the market for 60 days or less. This metric is crucial in identifying areas where real estate moves quickly, indicating high demand and investment viability. This methodology is designed to uncover the intricate patterns and trends that define emerging real estate hotspots, offering a nuanced understanding of regional market dynamics. The significance of this research is twofold. Firstly, it provides investors with crucial insights into areas with high growth potential, guiding investment strategies in the dynamic real estate landscape. Secondly, this analysis equips real estate professionals with knowledge about active markets which enables them to optimize their services and strategies to compete in the real estate market. The existing literature lays a foundation for our study.

Works such as "Predictive Modeling of U.S. Housing Prices Reveals Key Indicators of Real Estate Prices and Economic Health" and "Predicting Real Estate Prices in China: A Comprehensive Analysis and Model Development" offer insights into the factors influencing housing prices in different markets. The utilization of machine learning and data analytics in real estate is further exemplified in studies like "The Use of Machine Learning in Real Estate Research." Additionally, the impact of turbulent events, namely the COVID-19 pandemic, on housing preferences and market dynamics is explored in "The Impact of The COVID-19 Pandemic On the Demand for Density: Evidence from the U.S. Housing Market." Despite these extensive studies, there remains a gap in specifically identifying and characterizing the latest real estate hotspots in the U.S. through a comprehensive, data-driven approach. Our research aims to fill this gap, providing valuable insights for various stakeholders in the real estate sector. The outcomes of this study are not only beneficial for guiding investment decisions but also crucial for informing policymakers on aspects of land use, infrastructure, and city planning, thereby contributing to balanced and sustainable urban development.

2. Methods

A. Data Collection and Preparation

Our study utilized a comprehensive dataset from Zillow, comprising 40 CSV files. For our analysis, we selectively incorporated specific files relevant to our objectives. The datasets included were sale price, mean sale price, days on Zillow, region ID, region name, and the percentage of houses sold above the list price. This selection was crucial for identifying the characteristics defining real estate hotspots.

B. K-Means Clustering for Hotspot Identification

The initial phase of our analysis involved cluster identification within the data. To achieve this, we implemented K-Means Clustering, a suitable method for grouping data based on shared characteristics. Our approach commenced with Two-Dimensional Clustering, focusing on the mean sale price and days on Zillow. The data was merged with region identifiers, such as region ID and name, specifically targeting houses in individual cities. We established baseline clusters using data from a pre-pandemic period, January 2019. The optimal number of clusters was determined using the Elbow Method, which suggested four distinct clusters. This clustering process was then replicated for January 2021 and October 2023 data, allowing us to observe the evolution of clusters over time.

C. Linear and Logistic Regression Analysis

Following the clustering, we conducted linear regression analysis on the two-dimensional models. In this analysis, the days on Zillow served as the dependent variable, while the house price was the independent variable. This regression was applied to each cluster set (January 2019, 2021, and October 2023) to examine the relationship between market duration and housing prices.

D. Three-Dimensional Clustering

To deepen our analysis, we extended the clustering into three dimensions, incorporating the Percentage of Homes Sold Above List Price (PSA) alongside price and days on Zillow. Again, using the Elbow Method, we determined that four clusters were optimal for this model. This 3D clustering was executed to further understand the market dynamics. Additionally, we conducted a similar 3D clustering analysis focusing on the percentage of houses sold above the list price, using the same three parameters: price, days on Zillow, and PSA. Through these methodological approaches, our study aimed to capture a comprehensive and nuanced understanding of the real estate market's current state, particularly focusing on identifying emerging hotspots and understanding their defining characteristics.

3. Results

The results of our 2D clustering can be seen in *Figure 1* and *Figure 3*. The January 2019 Clustering sets a pre-pandemic baseline with four distinct clusters and only two containing

hotspots in the black and blue clusters. Cities in the upper green cluster represent many smaller rural cities such as Erie, PA, Boone, NC, and Glens Falls, NY. Cities in the middle red cluster are mostly rural and suburban areas such as Myrtle Beach, SC, Syracuse, NY, and Oklahoma City, OK. Cities in the lower blue cluster are mostly urban cities like Boston, MA, Miami, FL, and San Antonio, TX. Cities in the far right black cluster are the select uber-wealthy areas such as San Francisco, CA, Kahului, HI, and Glenwood Springs, CO. The January 2021 Clustering demonstrates where the housing market was at the height of the pandemic with four distinct clusters and three containing hotspots in the black, red, and blue clusters. Cities in the upper green cluster represent many smaller rural cities such as Barre, VT, Boone, NC, and Glens Falls, NY. Cities in the lower left black cluster contain mostly rural and suburban areas such as Erie PA, Syracuse, NY, and Oklahoma City, OK. Cities in the middle red cluster are mostly urban cities like Boston, MA, Miami, FL, and Denver, CO. Cities in the far right blue cluster are the select uber-wealthy areas such as San Francisco, CA, Kahului, HI, and Glenwood Springs, CO. The October 2023 cluster takes the same shape as the January 2019 Clustering with four distinct clusters and three containing hotspots in the green, blue, and black clusters. Cities in the upper red cluster represent many smaller rural cities such as Erie, PA, Boone, NC, and Glens Falls, NY. Cities in the middle green cluster are mostly rural and suburban areas such as Myrtle Beach, SC, Syracuse, NY, and Oklahoma City, OK. Cities in the lower blue cluster are mostly urban cities like Boston, MA, Miami, FL, and San Antonio, TX. Cities in the far right black cluster are the select uber-wealthy areas such as San Francisco, CA, Kahului, HI, and Glenwood Springs, CO.

The results of our linear regression to find hotspots can be seen in *Figure 5 and Figure 7*. The January 2019 regression has the equation $\text{Days on Zillow} = 91.87 - 3.46e-5 * \text{Price}$. The January 2021 regression has the equation $\text{Days on Zillow} = 54.9 - 1.048e-5 * \text{Price}$. The October 2023 regression has the equation $\text{Days on Zillow} = 37.55 + 1.264e-5 * \text{Price}$. The Logistic Regression results can be seen in Tables 1 to 3. Percentage Sold Above the List Price (PSA) remains a significant predictor of hotspots for all three years. The Intercept is significant in the first two logistic models. Price is only a significant predictor of hotspots in the 2019 model.

The results of our three-dimensional clustering models can be seen in Figures 8 to 10. The clustering of the 3D model is very similar to the 2D model. The January 2019 clustering has four distinct clusters and three contain hot spots in the yellow, purple, and green clusters. Cities in the upper right purple cluster represent many smaller rural cities such as Erie, PA, Jackson, MS, and Glens Falls, NY. Cities in the middle green cluster are mostly rural and suburban areas such as Charleston, SC, Syracuse, NY, and Oklahoma City, OK. Cities in the lower blue cluster are mostly urban cities like Miami, FL, San Antonio, TX, and New Orleans, LA. Cities in the far left yellow cluster are the select uber-wealthy areas such as San Francisco, CA, Los Angeles, CA, and Boston, MA. The January 2021 clustering has four distinct clusters and three contain hot spots in the yellow, blue, and green clusters. Cities in the upper right green cluster represent many smaller rural cities such as Erie, PA, Jackson, MS, and Glens Falls, NY. Cities in the middle yellow cluster are mostly rural and suburban areas such as Charleston, SC, Syracuse, NY, and Oklahoma City, OK. Cities in the lower purple cluster are mostly urban cities like Miami,

FL, San Antonio, TX, and New Orleans, LA. Cities in the far left blue cluster are the select uber-wealthy areas such as San Francisco, CA, Los Angeles, CA, and Boston, MA. The October 2023 clustering has four distinct clusters and all four contain hot spots. Cities in the upper right green cluster represent many smaller rural cities such as Erie, PA, Jackson, MS, and Glens Falls, NY. Cities in the middle yellow cluster are mostly rural and suburban areas such as Charleston, SC, Syracuse, NY, and Oklahoma City, OK. Cities in the lower purple cluster are mostly urban cities like Miami, FL, San Antonio, TX, and New Orleans, LA. Cities in the far left blue cluster are the select uber-wealthy areas such as San Francisco, CA, Los Angeles, CA, and Boston, MA.

4. Discussion

The aim of the study is to determine which states and regions are experiencing a surge in real estate investment interest and can be considered a “hotspot” for real estate opportunities. The two-dimensional clustering illustrated in *Figure 1*, *Figure 2*, and *Figure 3* reflects the real estate market's evolution over several distinct time periods. In January 2019 (*Figure 1*), the clustering depicts a clear demarcation of regions, with each cluster representing a unique combination of house prices and days on Zillow. This distribution suggests that at the beginning of 2019, market segments were well-defined, potentially reflecting stable and predictable market conditions. Moving to January 2021 (*Figure 2*), the clusters display a shift in market characteristics. The dispersion of data points indicates changes in the correlation between prices and days on the market, potentially as a result of the economic and social disruptions experienced globally. The distinctions between clusters may reflect varied regional responses to these disruptions, with some areas showing resilience or even growth, while others may have experienced declines in property values or increased time-on-market. The October 2023 clustering (*Figure 3*) reveals a further evolution of the market. The clusters in this period may represent a new real estate landscape shaped by post-pandemic economic recovery, shifts in work and living patterns, and possibly changes in consumer confidence and investment strategies. The spread and overlap of clusters could indicate a market that is less segmented than in previous years, suggesting a convergence of market characteristics across different regions or a general market trend that overrides regional differences.

The linear regression analyses presented in *Figure 5* and *Figure 6* illustrate an intriguing shift in the real estate market dynamics over time. Initially, as depicted in *Figure 5*, there is a negative correlation between house prices and days on Zillow, suggesting that properties lingering longer on the market tend to decrease in price. This is a common market behavior, where urgency or the lack thereof influences the transaction value. A similar trend is observed in *Figure 6*, where, despite a different time frame, the relationship between price and days on the market remains consistently negative. In general, a negative correlation suggests a buyer's market where there was more supply and demand of houses which led to price reductions over time to attract buyers. However, *Figure 7*, representing the October 2023 data, marks a notable

departure from this trend. Here, we see an unexpected positive correlation, indicating that house prices increase with longer market durations. This suggests a shift in seller behavior, where sellers are willing to wait longer periods to secure a sale at or above their desired price point. The real estate dynamics shifted post-pandemic from a buyers' market to a sellers' market. The pandemic influenced housing preferences and patterns which led to increased demand for certain property types. For example, after the pandemic, people started moving away from amenity-rich, urban centers and moved towards homes in less crowded neighborhoods with bigger spatial arrangements that were previously not in high demand. This may suggest why homes listed longer on Zillow started seeing high prices, as buyers were willing to pay more for a home that better fit the criteria buyers were looking for. Several factors could contribute to the shift in market dynamics. First, a robust economic outlook can lead to sellers believing that they can afford to wait for the right buyer who is willing to meet their price. Second, a limited supply of homes could lead to less pressure on sellers to lower prices, allowing for longer listing periods. Finally, shifts in buyer preferences, such as the trend towards remote work reducing the necessity to work in the office, may increase competition and willingness to pay a premium price for select properties.

The 3D clustering, incorporating the PSA, adds another layer to our understanding, visually depicting a relationship between the days on Zillow, price, and the PSA. This triad of factors provides a richer, more nuanced picture of the market, suggesting that not just price and time on the market, but also the broader economic context as encapsulated by the PSA, interact in complex ways to influence real estate dynamics. The observed shift in the correlation between market duration and price indicates a maturing market where sellers have more leverage and buyers' preferences may be evolving. Understanding these subtle yet significant changes is critical for stakeholders to navigate the real estate landscape effectively. Future research should aim to unravel these complexities further, providing clarity on the driving forces behind these market trends.

5. Conclusion

In this study, we explored the rapidly changing areas in the U.S. real estate market to find the best places for investment. Our research has identified several key real estate hotspots across the U.S., with high potential for investment. Areas such as Boston, MA, Miami, FL, and San Antonio, TX have emerged as urban centers where properties sell quickly, often within 60 days, indicating robust demand. Additionally, uber-wealthy regions like San Francisco, CA, Kahului, HI, and Glenwood Springs, CO, are notable for their high investment potential. Our analysis, using advanced data analytics and statistical analysis, revealed that post-COVID-19, these markets have seen a significant shift, with properties commanding higher prices even when listed longer. This trend is crucial for investors and real estate professionals, as it highlights evolving

market dynamics and pinpoints areas ripe for investment. By identifying these hotspots, our study provides strategic insights into the U.S. real estate market, aiding stakeholders in making informed decisions.

6. Appendix

Figure 1: January 2019 2D Clustering

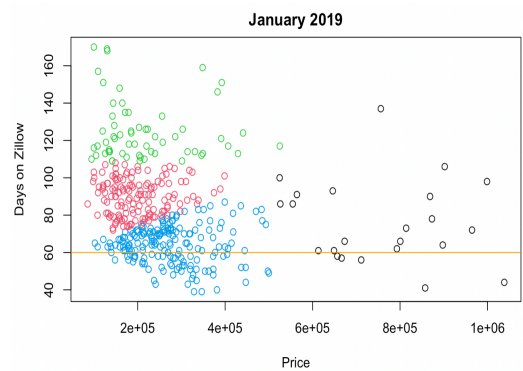


Figure 2: January 2021 2D Clustering

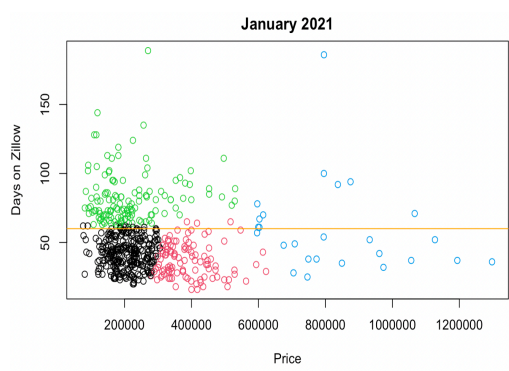


Figure 3: October 2D Clustering

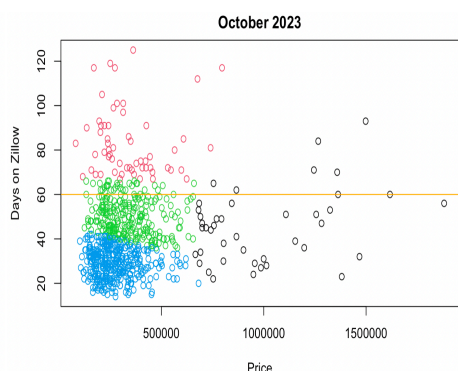


Figure 4: Elbow Method Plot to Find the Number of Clusters

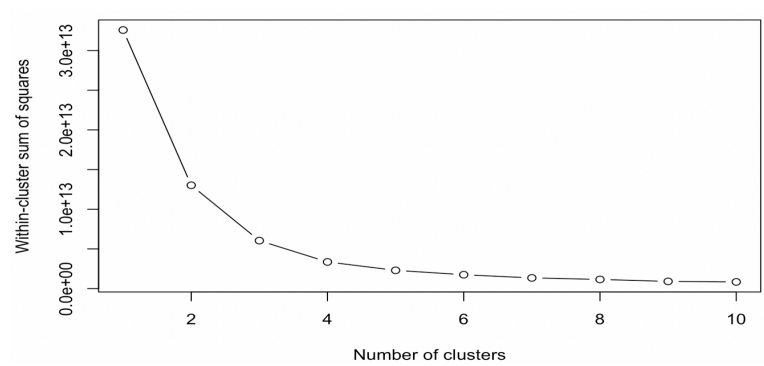


Figure 5: January 2019 Logistic Regression

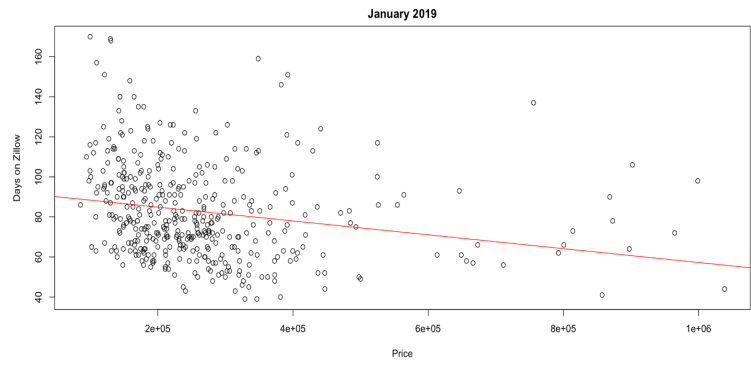


Figure 6: January 2021 Linear Regression

Figure 7: October 2023 Linear Regression

Figure 8: January 2019 3D Clustering

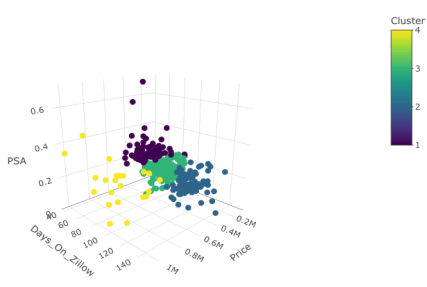


Figure 9: January 21 3D Clustering

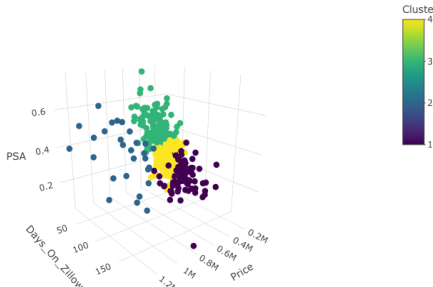
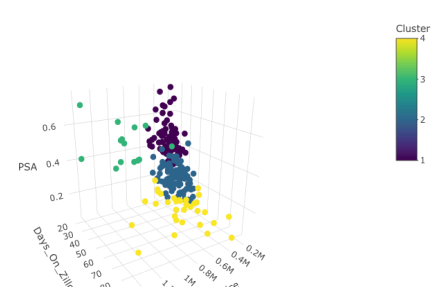


Figure 10: October 2023 3D Clustering



Logistic Regression Tables:

Table 1: January 2019

Coefficients	Estimate	Std. Error	Z-value	PR(> z)
Intercept	-4.988	.5911	-8.44	<2e-16***
Price	2.439e-06	9.661e-07	2.524	.0116*
PSA	15.37	2.36	6.512	.7.4e-11***

Table 2: January 2021

Coefficients	Estimate	Std. Error	Z-value	PR(> z)
Intercept	-1.865	.3813	-4.891	1e-06***
Price	-1.118e-06	8.012e-07	-1.395	.163
PSA	13.04	1.509	8.646	<2e-16***

Table 3: October 2023

Coefficients	Estimate	Std. Error	Z-value	PR(> z)
Intercept	-4.488	2.334	-1.923	.05452
Price	-3.600e-07	1.852e-06	-.194	.84591
PSA	48.4	15.92	3.041	.00236 **

7. References

1. “Housing Market 2024: Zillow Predicts US Home Prices Will Jump 6.5% by Next Summer.” *Yahoo! Finance*, Yahoo!, finance.yahoo.com/news/housing-market-2024-zillow-predicts-185404994.html. Accessed 11 Dec. 2023.
2. “Housing Market 2024: Zillow Predicts US Home Prices Will Jump 6.5% by Next Summer.” *Yahoo! Finance*, Yahoo!, finance.yahoo.com/news/housing-market-2024-zillow-predicts-185404994.html. Accessed 11 Dec. 2023.
3. “Housing Market 2024: Zillow Predicts US Home Prices Will Jump 6.5% by Next Summer.” *Yahoo! Finance*, Yahoo!, finance.yahoo.com/news/housing-market-2024-zillow-predicts-185404994.html. Accessed 11 Dec. 2023.
4. News, Tiffany Hagler-Geard/Bloomberg. “Zillow Signals More Pain Ahead in Online Real Estate.” *The Wall Street Journal*, Dow Jones & Company, 16 Feb. 2023, www.wsj.com/articles/zillow-signals-more-pain-ahead-in-online-real-estate-89fd1b41.
5. Faris, Francesca. “Know Your Rights: Understanding Fair Housing Laws & Housing Discrimination.” *Zillow*, 25 Aug. 2023, www.zillow.com/learn/fair-housing-laws-discrimination/.
6. *U.S. Home Prices May Be down but These Markets Are Soaring*, realestate.usnews.com/real-estate/articles/us-home-prices-may-be-down-but-these-markets-are-still-soaring. Accessed 12 Dec. 2023.
7. Choy, Lennon H. T., and Winky K. O. Ho. “The Use of Machine Learning in Real Estate Research.” *MDPI*, Multidisciplinary Digital Publishing Institute, 25 Mar. 2023, www.mdpi.com/2073-445X/12/4/740.
8. Liu, Guangjie. “Research on Prediction and Analysis of Real Estate Market Based on the Multiple Linear Regression Model.” *Scientific Programming*, Hindawi, 9 May 2022, www.hindawi.com/journals/sp/2022/5750354/.
9. *Predictive Modeling of U.S. Housing Prices Reveals Key ... - IEEE Xplore*, ieeexplore.ieee.org/document/9275969/. Accessed 12 Dec. 2023.
10. Liu, Sitian, and Yichen Su. “The Impact of the COVID-19 Pandemic on the Demand for Density: Evidence from the U.S. Housing Market.” *Economics Letters*, U.S. National Library of Medicine, Oct. 2021, www.ncbi.nlm.nih.gov/pmc/articles/PMC8439176/#:~:text=We%20find%20that%20the%20pandemic%20reduced%20the%20housing%20demand%20in,dwindling%20attraction%20of%20consumption%20amenities.

